

## Post Midsem

We will look at determination of spot rates, expected forward rates (pure expectation theory)

### Theoretical Spot Rate Curve (Bootstrapping)

| Security | Type | Maturity | Semi-Annual Coupon | Annualized YTM | Face Value | Current price | Spot Rate |
|----------|------|----------|--------------------|----------------|------------|---------------|-----------|
| 1        | ZCB  | 0.5      | -                  | 5              | 100        | 97.561        | 5.1.      |
| 2        | ZCB  | 1.0      | -                  | 5.25           | 100        | 94.9471       | 5.251.    |
| 3        | CB   | 1.5      | 2.75               | 5.5            | 100        | 100           | 5.511.    |
| 4        | CB   | 2.0      | 2.875              | 5.75           | 100        | 100           | 5.5771.   |
| 5        | CB   | 2.5      | 3                  | 6.0            | 100        | 100           | 6.031.    |
| 6        | CB   | 3.0      | 3.125              | 6.25           | 100        | 100           | 6.301.    |

Price of CB having maturity 1.5 yrs =

$$100 = \frac{2.75}{\left(1 + \frac{0.05}{2}\right)^1} + \frac{2.75}{\left(1 + \frac{0.0525}{2}\right)^2} + \frac{102.75}{\left(1 + \frac{S_{1.5}}{2}\right)^3}$$

$$S_{1.5} = 0.0551 = 5.511.$$

Similarly we can find  $S_2, S_{2.5}, S_3$

Now calculating forward rates

The implied semi annual forward rates on  $M$ -year bonds purchased  $T$ -years from the present can be calculated using a formula

$$f_{m,t} = \left[ \frac{(1 + S_{m+T})^N}{(1 + S_t)^i} \right]^{\frac{1}{N-i}} - 1$$

$S$ : Semi annual spot rate

$M$ : Time to maturity

$T$ : Time period from present to forward rate (in years)

$N$ : Number of semi annual periods to  $M+T$

$i$ : Number " " " to  $T$

Q Find out implied forward rate on 6 months bond 6 months from now?

$M = 6$  months

$T = 6$  months

Ans

$$(1+S_1) = [(1+S_{0.5})(1+f_{0.5,0.5})]^{1/2} \quad S_{0.5} = S_0$$

$$f_{0.5,0.5} = \frac{(1+S_1)^2}{1+S_{0.5}} - 1$$

$$\begin{aligned} f_{0.5,0.5} &= \frac{(1 + \frac{0.0525}{2})^2}{1 + \frac{0.05}{2}} - 1 \\ &= 0.027502 \end{aligned}$$

Calculate implied forward rate on a 0.5 yrs bond purchased 1 year from now

$$f = \left[ \frac{(1 + \frac{0.0551}{2})^3}{(1 + \frac{0.0525}{2})^2} \right]^{\frac{1}{3-2}} - 1$$

$$= 0.03015 = 3.01\% \text{ Semi annual rate}$$

Duration: weighted average of bond's time period with weights being each time period's relative present value of its cash flow

$$D = \sum_{t=1}^n t \frac{PV(CF)}{P_0}$$

↓  
Macaulay's duration

Q

$m = 4 \text{ yrs}$  coupon = 9% PAR = 1000 yield = 10%.

| $t$ | $CF_t$ | $PV(CF_t)$               | $PV(CF_t) / P_b$ | $t \times PV(CF_t) / P_b$ |
|-----|--------|--------------------------|------------------|---------------------------|
| 1   | 90     | $90 / 1.1 = 81.818$      | $81.818 / 968.3$ | $0.0849 \times 1$         |
| 2   | 90     | $90 / 1.1^2 = 74.388$    | .                | $0.076823 \times 2$       |
| 3   | 90     | $90 / 1.1^3 = 67.618$    | .                | .                         |
| 4   | 1090   | $1090 / 1.1^4 = 744.485$ | .                | .                         |
|     |        | <u>968.3</u>             |                  | <u>3.52</u>               |

Duration is a time measure

Immunitization strategy  $\rightarrow$  investor is immunized from interest rate risk if liability period = duration of assets

Properties of duration

- (i) Captures market interest rate, size of coupon, maturity I.R risk can be managed
- (ii)  $\therefore$  change of bond price given a change in yield } Modified duration  
We can measure price volatility of bond

$$P = C \left[ \frac{1 - \frac{1}{(1+y)^n}}{y} \right] + \frac{F}{(1+y)^n}$$

Taking 1<sup>st</sup> order derivative

$$\frac{dp}{dy} = \frac{C}{y^2} \left[ 1 - \frac{1}{(1+y)^n} \right] + \frac{n(F - C/y)}{(1+y)^{n+1}}$$

$$\text{Modified duration} = \frac{\left| \frac{dp}{dy} \right|}{P}$$

Slope of yield curve  
Price of bond

$$\frac{\text{Macaulay's duration}}{(1+y)} = \text{Modified duration}$$

Q coupon = 9% Maturity = 10 yrs FV = 100 bond selling at PAR semi annual coupon payment

$$\frac{dp}{dy} = \frac{4.5}{(0.045)^2} \left[ 1 - \frac{1}{(1.045)^{20}} \right] + 20 \frac{(100 - 4.5/0.045)}{(1.045)^{20}}$$

$$p = 100$$

$$\text{Modified duration} = 13$$

$$\text{annualized modified duration} = 6.5$$

Modified duration  $\rightarrow$  slope of price yield curve divided by price of bond  $(dp/dy)$

Characteristics of duration

1. Coupon  $\downarrow$  duration  $\uparrow$
2. Longer term to maturity, duration  $\uparrow$
3. For a zero coupon bond, maculay's duration = maturity

Q

Time = 10 years

yield = 9%

selling at 100 duration = 6.5 yrs

Now interest rate is increased to 9.1%.

% change in price of bond?

% change in price of bond =

$$\frac{\% \Delta p}{\Delta y}$$

$$\% \Delta p = \text{duration} \times \Delta y$$

$$= 6.5 \times (0.0910 - 0.09) = 0.0065$$

$$\text{bond price (at 9.1\%)} = \sum_{t=1}^{20} \frac{4.5}{(1.0455)^t} + \frac{100}{(1.0455)^{20}} = 99.3527$$

$$\% \text{ change} = \frac{99.3524 - 100}{100} = -0.6476\%$$

② IR ↑ 200 bp

$$6.5 \times 0.02 = \frac{\Delta p}{p} \Rightarrow \frac{\Delta p}{p} \times 100 = 1.3\%$$

but from bond price = 11.95%

For large change in yield, duration will give, less accuracy in price volatility

whenever the yield change is greater than the accuracy of price volatility by duration goes down

If yield ↑, duration will overestimate the price changes

If yield ↓ " " underestimate " "

Convexity → measure of change in slope of price yield curve (2<sup>nd</sup> order derivative) → convexity  $\frac{\partial^2 P}{\partial y^2}$

→ measures curvature of price yield curve

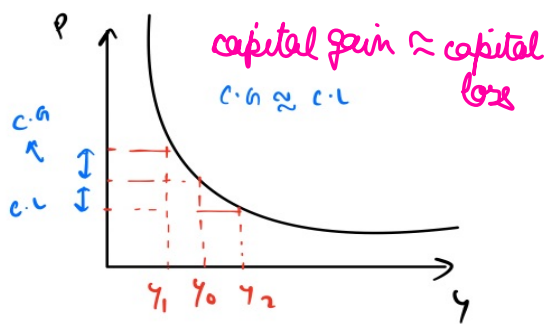
$$\text{Convexity} = \frac{\frac{2C}{y^3} \left[ 1 - \frac{1}{(1+y)^N} \right] - \frac{2CN}{y^2(1+y)^{N+1}} + \frac{N(N+1) \left[ F - \left( \frac{C}{y} \right) \right]}{(1+y)^{N+2}}}{P}$$

$$\text{annualized convexity} = \frac{\text{convexity}}{m^2}$$

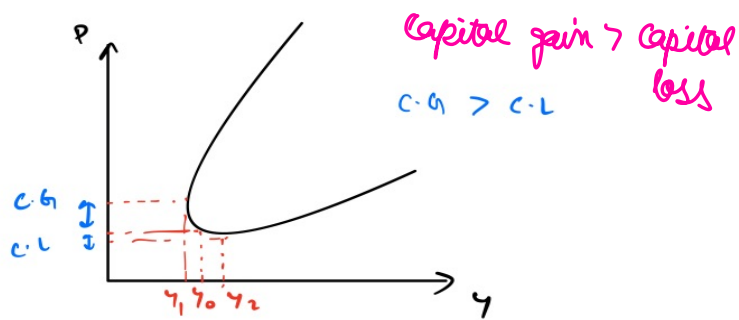
greater the bond's convexity, greater its capital gain and smaller its capital losses, for given absolute change in the yield

↳ asymmetrical gain & loss

If convexity of bond more → price performance of the bond more



(kind of symmetric)



(kind of asymmetric)

$$(ii) \quad \frac{\Delta P}{P} = \underbrace{\left( \text{modified duration} \right) \times \Delta y}_{\text{modified duration}} + \underbrace{\frac{1}{2} \text{ convexity} \times \Delta y^2}_{\text{convexity}}$$

(iii) If  $y \uparrow$ , convexity  $\downarrow$

(iv) For a given  $y$  & maturity, lower coupon, convexity  $\uparrow$ .

(v) For a given  $y$  & modified duration, lower coupon, convexity  $\downarrow$ .

(vi) Other things equal, greater the bonds convexity, the more valuable the bond is.

→ For a given yield & maturity lower coupon, greater the convexity

→ " " " & modified duration → lower the coupon

→ smaller the convexity  
greater the convexity, more valuable the bond is  
in terms of returns

5/10/23

## Bond Markets in India

### Government Security Market

1. Gov security bonds
2. Corporate bonds
3. PSU bonds → sometimes maybe classified as corporate

① Issued through auction process  
Maximum term to maturity is upto 30 years  
generally coupons are semi-annual

3 segments — Wholesale — banks, financial institutions, PDs, insurance, MFs  
Middle — companies, PFs, cooperative banks, NBFCs  
Retail — for individuals

SGH account → can be opened by PDs & commercial banks, insurance companies  
↳ opened with RBI  
(generally)

Subsidiary General Ledger

GILT account → opened with particular bank/entity having SGH account  
↳ entity will participate on behalf of you through SGH

for those who can't open SGH account

You open GILT account with our bank/PD

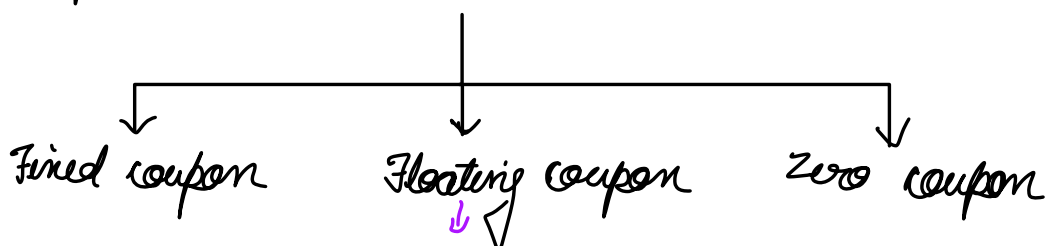
GILT → SGH → RBI

Bank/PD opens CSGH with RBI for our record & SGH for its own record

② Long term corporate bonds known as Debentures

Not as developed as ① in India

↳ not secured bonds (not backed by collateral)



related to benchmark, if benchmark change then coupon also change. eg:- interest rate

Single redemption bond → face value will be given after maturity

Multiple " " (Amortised bond) → periodically principal / par value will be given

② PSU bonds → Public Sector Undertaking → Ex. Bharat Petroleum, Hindustan, aeronautics Ltd.

Why unique?

→ PSU will decide kisko issue karna h bond

↳ Mostly privately placed → not issued to general public

↓  
sold to targeted investors

↓  
existing shareholders will be getting (generally)

→ Maturity period 5-10 years → issued by non-governmental but kinda backed by the government

→ Also called as Quasi Sovereign bonds

→ Some of PSU bonds are not tradeable (unlike corporate bonds)

## Stock Market

- various matters

- stock valuation — DCF models

— Relative valuation models

— Contingent claim models → for derivatives (futures, options, swaps)

→ Performance of IPO?

1. Primary market → how IPO pricing done? Why companies go for IPO?

2. Secondary market

↳ common stocks, how different indices are constructed?

Market microstructure of market → listing, orders, Market orders etc.

Stocks → common share  
→ preferential share

## Common

1. voting rights
2. Dividend is not fixed (May not give)
- 3.

## Preference share

- No voting rights  
Dividend is fixed (mandatory)  
If company is liquidated preference order  
1. Bonds 2. Preference 3. Common

## Two type of preference shares

- ① Cumulative → if for some reason company couldn't give dividend, then company will give next year along with pending last year's dividend

Non-Cumulative - Won't get dividend next year, if company couldn't give dividends

②

Convertible - after a point of time, preference share can be converted to common after a particular point of time

Non Convertible - can't be converted to common stock after particular point of time

- ③ Redeemable - matures after a fixed period of time, also considered as debt instruments

Non Redeemable -

- ④ Participating - chance of getting high dividend if company's profit increased or goes above a certain value

Non participating -

net worth / no. of outstanding shares

Share has a face value, book value, market value

it is the minimum value of the share what the company has to disclose at IPO

another proxy → intrinsic value (discounted cash flows)

overvalued  $\rightarrow$  should sell  
undervalued  $\rightarrow$  " buy

If market value  $>$  book value (Overvalued)

(Market value  $<$  Face Value)

Face Value  $\rightarrow$  if at any point of time market value is going down as compared to face value then company become "hopeless".

Face value  $\rightarrow$  no such model, formula to get  
Random numbers like ₹5, ₹10

## ② Valuation of Equity 6/10/23

### → Discount Flow Approach

we use 3 cash flows

- (i) Dividend  $\xrightarrow{\hspace{2cm}}$   $r = \text{cost of equity}$
- (ii) operating cash flow  $\xrightarrow{\hspace{2cm}}$   $r = \text{cost of capital}$
- (iii) free cash flow  $\begin{cases} \rightarrow \text{equity } (r = \text{cost of equity}) \\ \rightarrow \text{firm } (r = \text{cost of capital}) \end{cases}$

Now using Dividend as a cash Flow, we need

- ① Dividend
- ② Growth rate of Dividend
- ③ Discount rate / required rate of return

$$\text{growth rate} = \text{Retention Rate} \times \text{ROE}$$

$$\downarrow$$
$$RR = 1 - \text{Dividend Payout Ratio}, \text{ROE} = \frac{\text{PAT}}{\text{Total Equity}}$$

To find cost of equity  $\rightarrow$  we use CAPM

$$\begin{array}{ccc} E(R) & = & \alpha + \beta (R_M - R_F) \\ \downarrow & & \downarrow \\ \text{cost of equity} & & \text{market risk premium} \end{array}$$

Dividend  $\Rightarrow$  proportion of earnings per share

$$\text{Rs } 50 \times 20\% = \text{Rs } 10$$
$$\downarrow$$

Dividend Payout Ratio.

### GORDON'S MODEL

- ① Growth rate of dividend is constant.
- ② Discount rate  $>$  Growth rate of Dividend ( $r > g$ )

$$P = \frac{D_0 (1+g)}{(r-g)} = \frac{D_1}{(r-g)}$$

If growth rate changes over time, this formula is not valid.  
∴ we need other models

- Two-Stage Model
- H-Model
- Three-Stage Model

### Two-Stage Model

- In first period company has a high growth rate.
  - After certain time growth rate goes down and remains stable over time.
- Life Cycle Theory  
of a company

$P_0$  = PV of stock in High Growth Period + PV of terminal price

$$P_0 = \frac{DPS_0 (1+g) \left[ 1 - \frac{(1+g)^n}{(1+r)^n} \right]}{(r-g)} + \frac{DPS_{n+1}}{(r-g_n)(1+r)^n}$$

$$P_0 = \sum_{t=1}^n \frac{DPS_t}{(1+r)^t} + \frac{P_n}{(1+r)^n}, \quad P_n = \frac{DPS_{n+1}}{r-g_n}$$

$DPS_0$  : current period dividend

$r$  : discount rate

$g$  : growth rate in initial period (first  $n$  years) extraordinary growth rate.

$g_n$  : steady period growth rate.

Ex:-  $EPS = 10.16$  ,  $g = 4\%$  , length of high growth period = 4 years  
payout ratio (high growth period) = 52%.

payout ratio (stable growth period) = 60%.

$r = 12\%$  , expected growth rate (stable period) = 3%.

$g_n$

$$DPS_{n+1} = \frac{10.16 \times (1.04)^4 \times 0.6 \times 1.03}{0.12 - 0.03} = \text{Rs } 81.62$$

$$P_0 = \frac{10.16 \times 0.52 \times 1.04 \left[ 1 - \frac{(1.04)^4}{(1.12)^4} \right]}{0.12 - 0.04} + \frac{81.62}{(1.12)^4}$$

$$= \text{Rs } 69.49$$

→ Difficult to specify time period of supernormal growth

3 limitations of this model

(i) difficult to specify the period of supernormal growth. (time period) → sudden growth drop not possible

(ii) growth has suddenly dropped (Sudden change not possible)

(iii) Terminal price is discounted using Gordon's formula. So all limitations of Gordon's model are also included.

→ Gordon formula used  
↳ uski bhi limitations aaygi

#### H-model

Assumption: gradual change in growth rate  
dividend payout ratio is constant

$$P_0 = \frac{DPS_0 (1 + g_n)}{r - g_n} + \frac{DPS_0 \times H (g_a - g_n)}{(r - g_n)}$$

↓  
stable growth  
rate period

$g_a$ : initial growth rate

$g_n$ : growth rate at the end of 2H years which extends to the initial period.

(i) cost of equity may change during stable period

(ii)  $g_a - g_n$ : growth rate is changing equally over time. so we take avg. by dividing by 2. (∵ H)

Ex:-

$$EPS_0 = \text{Rs } 2$$

$$\text{payout ratio} = 50\%$$

$$g_a = 18\%$$

growth rate was expected to decline at the rate of 2% / year over the next 6 years to make  $g_n = 6\%$

$$\text{required rate of return} = 13\%$$

Sol'n

$$P_0 = \frac{2 \times 0.5 (1 + 0.06)}{(0.13 - 0.06)} + \frac{2 \times 0.5 \times 3 \times (0.18 - 0.06)}{(0.13 - 0.06)}$$

$$P_0 = 15.14 + 5.142$$

$$P_0 = \text{Rs } \underline{\underline{20.28}}$$

11/10 Three phases of growth of a company

High growth  $\rightarrow$  low payout ratio  
 Transition  $\rightarrow$  higher  
 Stable  $\rightarrow$  highest

Discount rate / cost of equity  $\rightarrow$  changes

Q Let initial growth rate = 19.38%.

Current EPS = 33.27

Payout Ratio = 16.53%.

Initial cost of equity = 13.7%.

Stable growth rate = 4%.

Stable payout ratio = 65.22%.

" COE = 11.5%.

High growth period = 5 years

here we have assumed linear decline or growth

| Year | EPS                                | Growth rate                            | Payout | EPS $\times$ payout<br>$\rightarrow$ DPS | COE   | Discounting                      |
|------|------------------------------------|----------------------------------------|--------|------------------------------------------|-------|----------------------------------|
| 1    | 39.72                              | 19.38                                  | 16.53% | 5.5                                      | 13.7  | 1.137                            |
| 2    | 47.41                              | .                                      | .      | 6.57                                     | .     | $(1.137)^2$                      |
| 3    | 56.6                               | .                                      | .      | 7.34                                     | .     | $(1.137)^3$                      |
| 4    | 67.57                              | .                                      | .      | 9.36                                     | .     | .                                |
| 5    | 80.66                              | 19.38                                  | 16.53% | 11.17                                    | .     | .                                |
|      |                                    |                                        |        | 13.34                                    | 13.7  | $(1.137)^5$                      |
| 6    | $80.66 \times 1.16$<br>= 93.82     | $(19.38 - 4) / 5 = 3.076\%$<br>16.075% | 26.27% | 24.64                                    | 13.26 | $(1.137)^5 (1.1326) =$<br>2.1527 |
| 7    | $93.82 \times 1.323$<br>= 106.22   | 13.23%                                 | 36.91% | 38.25                                    | 12.82 | $(2.1527) (1.282) =$<br>2.4281   |
| 8    | $106.22 \times 1.1015$<br>= 117.06 | 10.15%                                 | 45.47% | 53.52                                    | 12.38 | 2.7267                           |
| 9    | $117.06 \times 1.0708$<br>= 125.29 | 7.08%                                  | 55.48% | 69.5                                     | 11.94 | 3.052                            |
| 10   | 130.3                              | 4%                                     | 65.22% | 84.98                                    | 11.5  | 3.403                            |

$$\left. \begin{array}{l} 6.57 / 1.137 = 5.77 \\ 7.81 / 1.2928 = 6.06 \end{array} \right\} \text{Discounting DPS}$$

6.37

6.68

7.2

31.90

For first five  
year

For transition phase  $\rightarrow$  94.53 (sum of discounted DPS)

(next 5 years)

$$\text{Terminal price} = \frac{130.3 (1.04) (0.6522)^5}{0.115 - 0.04} = 1178.41$$

$$\frac{1178.41}{3.40} = 345.99$$

$$\text{Value of stock} = 345.99 + 31.90 + 94.53 = 472.42$$

/ Intrinsic value

$\hookrightarrow$  when using a 3 stage model

$\hookrightarrow$  the tedious task is transition  
phase  $\rightarrow$  have to do manually

$\downarrow$   
for high growth & stable  
we have formulas

Relative Valuation

we use multiples:-

1. P/E ratio (most popular)
2. P/B (price to book ratio)  
↓  
book value
3. P/S (price to sales ratio)

within some industry, which company is best

$$P_0 = \frac{DPS_1}{r - g_m}$$

$$\underbrace{P_0/E_0}_{\text{price earnings ratio}} = \frac{\underbrace{DPS_1/E_0}_{\text{payout ratio}}}{(r - g_m)}$$

→ relationship b/w PE &amp; constant growth

$$\frac{P_0}{E_0} = \frac{\text{payout ratio} (1 + g_m)}{(r - g_m)}$$

→ if you are using current earnings ( $E_0$ )

$$\frac{P_0}{E_1} = \frac{\text{payout ratio}}{r - g_m}$$

→ if using expected earnings per share

$$P_0 = \frac{E_0 (\text{payout ratio}) (1 + g) \left[ 1 - \frac{(1 + g)^n}{(1 + r)^n} \right]}{r - g}$$

$$+ \frac{E_0 (\text{payout ratio}) (1 + g)^n (1 + g_m)}{(r - g_m) (1 + r)^n}$$

$$\text{EPS} = 1.5$$

growth rate = 10% (first 5 years)

payout ratio = 20% (first " ") after 5 years = 50% (payout ratio)

Stable growth rate = 5%

Cost of equity = 12%

Find P/E

$$\frac{P}{E} = \frac{(0.2)(1.1) \left[ \frac{1 - \frac{(1.1)^5}{(1.12)^5}}{0.12 - 0.1} \right] + \frac{(0.5)(1.1)(1.05)}{(0.12 - 0.05)(1.12)^5}}{1}$$
$$= 7.794$$

→ you have to invest 7.794 for 1 earnings

↓  
so lower P/E is better for investor

### Limitations of P/E

→ Earnings can be -ve

### P/B ratio

$$P_0 = \frac{DPS_1}{r - g_m} = \frac{DPS_0(1+g_m)}{r - g_m}$$
$$= \frac{EPS_0 \times (\text{payout ratio}) \times (1+g_m)}{r - g_m}$$

EPS = ROE × Book value of equity

$$P_0 = \frac{ROE \times BV_0 \times \text{payout ratio} \times (1+g_m)}{r - g_m}$$

$$\frac{P_0}{BV_0} = \frac{ROE \times (\text{payout ratio}) \times (1+g_m)}{r - g_m}$$

→ *conversion*

$$g = \frac{\text{Retention ratio} \times ROE}{1 - \text{payout ratio}}$$

$$\text{payout ratio} = \left( 1 - \frac{g}{ROE} \right)$$

if the company doesn't pay dividends even then we can calculate P/B ratio

add formula

- P/B ratio may not be useful for companies which have very less tangible assets (fixed assets) → book value kam hota, P/B may be misleading
- P/B ratio may not be beneficial if earnings are -ve

P/S ratio

$$P_0 = \frac{DPS_1}{r - g_m} = \frac{DPS_0(1+g_m)}{r - g_m}$$
$$= \frac{EPS_0 \times (\text{payout ratio}) (1+g_m)}{r - g_m}$$

$$NPM = \frac{EPS}{\text{sales per share}}$$

$$P_0 = \frac{\text{sales per share} \times NPM \times \text{payout ratio} \times (1+g_m)}{r - g_m}$$

$$\frac{P_0}{\text{sales per share}} = \frac{NPM \times \text{payout ratio} \times (1+g_m)}{r - g_m}$$

$$g = \text{retention ratio} \times ROE$$

$$\underbrace{\frac{\text{net profit}}{\text{sales}}}_{NPM} \times \frac{\text{sales}}{BV \text{ of equity}}$$

$$NPM = g \times \frac{BV \text{ of equity}}{\text{sales}} \times \frac{1}{RR}$$

## Factors affecting P/S

- growth
- BV of equity
- sales
- retention ratio
- cost of equity

P/S → ① can't be -ve ② not influenced by accounting methods used for calculation of depreciation

advantages

→ Free cash flow to equity



can be used in place of dividends for equity valuation

Free cash flow = Net income - (CAPEX - Depreciation) - change in working capital  
+ (Net debt raised - Debt repayment)



can be used in place of dividends

Two cash flows used for valuation of equity — free cash flow  
— dividends

Investors mostly use P/E ratio



nature of business should be same



same industry



compare with industry average

Companies have low P/E, P/B, P/S → value stocks

" " high " " " → growth stocks

Market

## Primary market instrument

→ IPO

→ FPO

→ Private placement

→ Bonus issues

→ Right issues

→ FPO

- Follow up public offering

- To raise capital after IPO

→ Right Issue

- company issuing shares only to the existing shareholders at a particular date

↳ record date

→ Bonus Issue

- giving share as a bonus to the existing shareholders

→ Private placement

- Issued to specific investors, should not cross 49, excluding QIBs &amp; employees

↳ in one go

↳ Qualified institutional Buyers

- In a particular year, this should not exceed 200

## → Different ways of issuing IPO

### 1. Profitability Route - Entry norm 1

- Net tangible asset of company should be atleast 3 crores in each of previous 3 years
- Net worth of at least 1 crore in preceding 3 years
- Company's profit should be +ve in preceding 3 years
- If company changed name in past 3 year, then 50% of revenue should come from new entity company has formed
- Issue size of share  $\leq 5 \times$  (Net worth as per last financial year)

### 2. Entry Norm - 2: QIB route

- At least 50% IPO given to QIBs
- Min post issue value  $\geq 100\text{cr}$ .

### 3. Entry Norm 3: Appraisal Route

- 15% of IPO by financial institutions
- Min post issue value  $\geq 100\text{cr}$

## → Price of IPO

### • Fixed price process

- Before issuance, price of IPO fixed by issuer
- 100% advance payment required

- 50% IPO reserved for small investors ( $\leq 1$  lakhs)
- Price approved by SEBI

### • Book building process

- Based on supply demand
- company appoint Book Runner Lead Manager (BRLM) i.e. Investment Bank
- BRLM makes a draft prospectus, which discloses all major information about the company
- sent to SEBI for approval and certified by SEBI
- After prospectus is ready, disclosed to public
- BRLM fixes a bid price, but is not disclosed to public
- BRLM appoints syndicate members (banks) & tentative price is disclosed to them
- They do a "roadshow" and provide info. about the company
- Demand is gauged through this and price range is decided
  - ↓
- 20% price range given to public & public can  $\pm 20\%$  bid in this price range
- Final price (Market clearing price) decided
- Quoter checked (QIB, Retail Investor etc.)
- Info given to SEBI, final allocation completed
- QIB  $\rightarrow$  only 10% advance payment required
- Retail investor  $\rightarrow$  100% advance required
- Once IPO done, listing completes and instrument enters secondary market

But

why companies go public??

1. Value based hypothesis

- Raise capital
- Diversify risk
- Rebalance debt equity ratio
- Increase liquidity
- Increase monitoring capacity
- Increase brand name / publicity

Cost of IPO

- loss of confidentiality
- high initial expenses
- Dilution of control
- Fear of litigation

→ Price of IPO offer, listing price (after IPO) and closing price > offer price



the short term  
performance



under pricing of IPO

Reasons / factors  
affecting IPO performance ←

# 1. Company specific

- company size
- " age
- Profitability
- Institutional ownership
- backed by VC? Previous funding

# 2. Issue specific

- offer price
- listing delay
- subscription rate
- underwriter reputation

# 3. Market specific

- Market Timing
- Market Volatility
- Macroeconomic business cycle
- Uncertainty in market
  - ↳ inflation / oil price change

18/10/23

Stock market  $\rightarrow$  regulated by SEBI

$\hookrightarrow$  demand & supply will decide price

$\rightarrow$  zero sum game

Benchmark  $\rightarrow$  general practice is compare return with market index

Two strategies

Active

- Always wants to beat the market
- get more return than market

Passive investor

- minimise the risk, follow the market

Market Index - judge overall market performance

How market index is constructed

- 4 inputs required
- Weights
  - Sample (have to choose out of population)
  - Base year
  - Base value

for any type of index, including stock market

What should be an ideal sample?

- $\rightarrow$  Should consider breadth of the market
- $\rightarrow$  Representative across industries
- $\rightarrow$  should represent population

BSE - SENSEX  
NSE - NIFTY

Every stock exchange has its own criteria, generally large companies are part of Index in case of India covering 50-60%. Market capitalisation

## Base Year

- should be a normal year (2020-2021  $\rightarrow$  can't be base year)
- where you do not have bigger natural calamity or political instability

Base value — was 100 for BSE  
1000 for NSE

Weights — 1) Equal weights — can be given on basis of price, market value  
2) Pure weight  
3) Value weight  
4) Free float value weight

Ex 3 stocks A, B, C chosen for index

|                 |    |               |
|-----------------|----|---------------|
| A — 60,000      | BP | current price |
| B — 20,000      | 30 | 45            |
| C — 90,000      | 25 | 80            |
| <u>        </u> | 65 | 85            |
| 05 shares       |    |               |

We always consider closing price

Let base value = 100

If wt given on basis of price — more price, more weight

$$\text{Equal wt.} = \frac{1}{3} \left[ \frac{45}{30} + \frac{80}{25} + \frac{85}{65} \right]$$
$$= 2.0033$$

$$100 \times 2.0033 = 200.33$$

$$\text{Price weight multiplier} = \frac{45+80+85}{30+25+65} = 1.75$$

$$\text{Index value} = 175$$

$$\text{Value weight} = \frac{45 \times 60k + 80 \times 20k + 85 \times 90k}{30 \times 60k + 25 \times 20k + 65 \times 90k} = 1.466$$

$$\text{Index value} = 146$$

|             |             |                              |
|-------------|-------------|------------------------------|
|             |             | % change                     |
| 30 × 60,000 | 45 × 60,000 | 50                           |
| 25 × 20,000 | 80 × 20,000 | 220                          |
|             |             | 30.76                        |
|             |             | <u>100.25</u> → avg % change |

$$\text{Index value} = 200.25 \quad (\text{Market value basis})$$

Free flow shares → Available with public + frequently traded

↳ generally excludes shares held by directors, founders, government, group companies, shares held by employee welfare trust, promoters

Free flow factor → 0.6 → 60% of total shares are frequently traded

↳ Rounded off to nearest 0.5 or divisible by 5

0.43 → 0.45 (Rounded off)

0.41 → 0.40

BSE, NSE → used market value weight upto 2003

↓

Now using free flow market capitalisation

→ Formula will remain same, just multiply free flow factor

Base year for BSE  $\rightarrow$  1975

NSE  $\rightarrow$  1993-1994

$\hookrightarrow$  established in 1992

19/10/23

## Market Microstructure

1. Listings
2. Security groupings → quality of stocks
3. Orders
4. Margin Trading
5. Circuit breakers

BSE, NSE has made diff. type of groupings

BSE - 4 types of security groupings - A, B, T and Z

**A category stocks** - companies should be registered atleast 3 months before in stock market

✓ imp ✓

stocks should be traded in secondary market in min. of 98% trading days in last 3 months  
↳ highly liquid

✓ imp ✓

companies with min. non promoter holding of 10%.

Promoter - Ambani for Reliance etc. Murthy for Infosys

have to take permission from exchange

↳ can't be traded freely

**T category** - settled on trade to trade basis and they are under surveillance

**Z category** - Investor's grievances, complaints are not being addressed by company  
- Over the period of time company has failed listing requirement  
- May not have made required arrangements with both

the depositories like CDSL, NSDL for dematerialisation of their securities

B category - companies neither in A, T, Z category

For NSE classified on basis of liquidity and impact cost

- Stock which are traded at least 80% of days for previous 6 months constitute group I and group II

- For stocks traded at least 80% of days with impact cost less than or equal to 1%  $\rightarrow$  group I

> 1% impact cost - group II

group III - not traded for 80% of days

Impact cost - marking out percentage price movement caused by order size of Rs 1 lakh from average price of best bid and offer price in the order book

Let best bid price be ₹20 (average)

agar 1 lakh ka stock khareid lie, so let price increased to ₹20.1 if that transaction has effect on price impact

be < 1% then group I

Trade effect / impact cost will be more for less liquid stocks

liquidity - how frequently stock traded & bid-ask spread

Trading system — order driven  
— quote driven

Order driven - displays all bid and ask prices, whoever is interested  
1  
Indian market  
- no guarantee your trade will execute or not

quote driven - only bid and ask prices of market makers are displayed

↓

US follows

↓

don't interest for  $\leftarrow$  dealers  
themselves  $\rightarrow$  big players / financial institutions

Market order - best price at a particular time

limit order

stop-loss order

Market if touched order

Market not held order

Market order -

at the best price currently  
available in the market

limit order - Instruction to trade at best price only if it is less than limit price specified by investor

Stop loss order - it is the stop instruction <sup>which stops or order</sup> from executing until stock price reaches price specified by investor

- can minimise your loss

Market if touch order - It is a market order which is actually when price reaches some present

Market not held order - these orders where brokers don't need to fill immediately, traders use their order instruction

- Broker has the discretion to execute the trade on behalf of trader

#### ④ Margin Trading

- Actual Margin
- Initial Margin
- Maintenance margin

when actual margins fall below maintenance margin

Margin call

Initial margin - money which you have put

Maintenance margin - at any point

$$10,000 - 5,000 + 5,000$$

Stock price - let now 80  $\rightarrow$

8,000 value  
- 5000 broker  
3000 your share

$$\frac{3000}{8000} = 37.5\% \rightarrow 730\% \text{ (Margin call)}$$

$$\text{Marginal call price} = \frac{\text{Amount of money borrowed}}{\text{No. of shares (1 - maintenance margin percentage)}}$$

$$= \frac{5000}{100(1-0.3)} = 71.43$$

for price 71.43 se niche aaga  $\rightarrow$  you will get margin call

Short selling - first you sell, then you buy

borrow from broker, sell at high price, buy at low price, and then return

If index value fluctuate by 10% before 1 pm  $\rightarrow$  then market trading will be halted for 45 mins

If 19/11 1-2:30pm - halted for 15 mins

" after 2:30pm - not halted

If index fluctuate 15% before 1 pm - 1 hr 45 mins halted

1-2:30pm - 45 mins halted

after 2:30pm - Remaining day

20%  $\rightarrow$  anytime during market hours - Remainder of the day

stock fluctuation - when large volume of stock  $\rightarrow$  bought or sold

30, 31 Oct  
Mon, Tue Online classes  $\rightarrow$

11/11/23

## Derivative

↳ main objective - hedging the risk

4 type of derivative contract

1. Forward
2. Futures
3. Options
4. Swaps

- to change nature of asset & liabilities → swaps
- exploit arbitrage opportunities → when law of one price is violated

### Forward

- Private contract b/w 2 parties
- Not standardised
- Chance of credit risk
- customised
- OTC
- specific delivery date
- Always settled at maturity

### Futures

→ special type of forwards

- standardised
- Traded on exchange
- No fixed delivery date (range of date) delivery can be done beforehand
- Mostly closed before maturity
- Practically no credit risk

### Options

- Buyer has the right but does not have the obligation to exercise the contract
- option buyer pays a premium to option seller to enter the contract

### Futures

- No premium

Open Interest: total no. of outstanding contracts that are held by market participants at the end of the trading day

Implied interest rate: cost of carry or the total cost required to carry a commodity

Transaction, storage, insurance cost  $\rightarrow$  For commodities  
Transport cost

Cost of financing - dividends  
 $\downarrow$   
risk free rate

} financial assets

Implied Volatility - measured through option pricing model, you need other inputs, once put you get implied volatility

Basis - diff. b/w spot price & future price  
determined

Contango - market condition when future price is above the expected future spot price

Backwardation - " " " below "

opposite of contango

Strike price - exercise price, pre-decided price at which option contract can be exercised

Daily Settlement price - closing price of future contract for the trading day  
- average price of last half an hour trading price

## Option premium

Expiration date - date on which contract is exercised

→ min price fluctuation

Tick size, lot size, daily price limit

## Components of Derivative contract

- lot size
- Tick size
- Daily price limit - restriction of price movement in a single date
- Expiry date
- Option Premium
- Delivery date & place

Spread - diff. b/w two future price, if prices are of same underlying good with diff. expiration date then spread is called → intra commodity spread

if price are of diff. underlying good with same / diff. expiration date then spread → inter commodity spread

NPT EL

derivatives



financial instrument whose payoff depends on  
a fundamental good Ex → grains, coffee, silver, bonds

Payoff of the contract → depends on price of the fundamental good

Use of derivatives

- To hedge risks
- To speculate
- To change nature of liability
- To look in arbitrage profit

Derivative instruments

4 type of derivative contract

1. Forward
2. Futures
3. Options
4. Swaps

Forward

→ agreement / private contract b/w 2 parties

- Private contract b/w 2 parties
- Not standardised
- Chance of credit risk
- Customised
- OTC
- Specific delivery date
- Always settled at maturity

- specified delivery date

Futures

→ special type of forwards → standardised

- Standardised
- Traded on exchange
- No fixed delivery date (range of date) delivery can be done beforehand
- Mostly closed before maturity

- Practically no credit risk

Range of delivery date

- options
  - calls
  - puts

→ right to buy

Calls → gives buyer the right but not the obligation to buy a given quantity of underlying asset at a given price on or before given future date.

→ right to sell

Puts - gives seller the right but not the obligation to sell a given quantity of underlying asset at a given price on or before given date

- swaps → mostly designed to convert liability to asset & asset to liability

- Private agreement b/w two parties to exchange cash flows in future according to a prearranged formula (Regarded as portfolios of forward contracts)

→ If ↑, fresh funds are flowing in the market, if ↓ means market is liquidating  
Open Interest: total no. of outstanding contracts that are held  
by market participants at the end of the trading day

Implied interest rate: cost of carry or the total cost required to  
carry a commodity

Transaction, storage, insurance cost → For  
Transport cost commodities

Cost of financing - dividends  
↓  
risk free rate  
For Equity futures

Implied Volatility - measured through option pricing model, you need  
other inputs, once put you get implied volatility

Basis - diff. b/w spot price & future price  
determined

Contango - market condition when future price is above the expected  
future spot price

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opposite of contango

Strike price - exercise price, pre-decided price at which option contract  
can be exercised

Daily  
Settlement price - closing price of future contract for the trading day  
- average price of last half an hour trading price

Option premium - price paid by buyer to seller to acquire the right to buy or sell

Expiration date - date on which contract is exercised, on expiration date either the

→ min price fluctuation

Tick size, lot size, daily price limit

option is exercised or it expires worthless

### Components of Derivative contract

- lot size
- Tick size → minimum price fluctuation
- Daily price limit - restriction of price movement in a single date
- Expiry date
- Option Premium
- Delivery date & place

Spread - diff. b/w two future price, if prices are of same underlying good with diff. expiration date then spread is called → intra commodity spread

if price are of diff. underlying good with same / diff. expiration date then spread → inter commodity spread

- sum of all outstanding long & short future market positions is always equal to zero

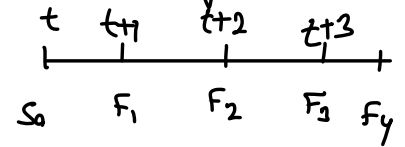
## Relationship b/w future price & spot price

Basis = Current spot price - future price

Normal market: Price for more distant future higher than nearby future

so future price > spot price

Inverted market: Future price < spot price



Convergence  $\rightarrow$  at time of expiration  $\rightarrow$  future price & spot price equal

$\downarrow$   
so basis becomes zero

$$S_0 < F_1 < F_2 < F_3 < F_4$$

$\downarrow$   
why because of risk

Spread  $\rightarrow$  diff. b/w two future prices

$\downarrow$   
if two future prices are of same underlying good but with diff. expiration dates  $\rightarrow$  spread is known as intra commodity spread

" " " " of diff. underlying goods with diff or same expiration dates  $\rightarrow$  inter commodity spread

## options

American option  $\rightarrow$  can be exercised at any time during its life

European "  $\rightarrow$  exercised only at maturity

Buyer of option (long posn)

Writer of option (short posn)

Call

Right to buy asset

Obligation to sell asset

Put

" " sell asset

" " buy "

# Concept of Money ness

| Condition                            | Call Option         | Put Option          |
|--------------------------------------|---------------------|---------------------|
| $S_0 > E \rightarrow$ exercise price | In-the-Money        | Out-of-the Money    |
| $S_0 < E$                            | Out-of-the Money    | In-the-Money        |
| $S_0 = E$                            | <u>At-the-Money</u> | <u>At-the-Money</u> |

market price on that day

In the money  $\rightarrow$  it will be exercised  
Out of the money  $\rightarrow$  it won't be exercised

Intrinsic value  $\rightarrow$  amount by which it is in the money

## Intrinsic Value and Time Value

- Option Premium = Intrinsic Value (Parity Value) + Time value (Premium over parity)
- Intrinsic value refers to the amount by which it is in-the-money
- Option which is out-of-the money has a zero intrinsic values
- For a call option which is in the money, the intrinsic value is the excess of stock price over the exercise price
- For a put option which is in the money, the intrinsic value is the excess of exercise price over the stock price

Option premium  
/ \  
Intrinsic + Time  
value value

Market value on that day - strike price

Pricing

$\rightarrow$  security providing no income  
 $\rightarrow$  " " a given amount of income  
 $\rightarrow$  " " a known yield

} three types

1. security providing no income

$$F = S(1+k)^T$$

$\downarrow$   
no income in between

$F \rightarrow$  future price

$S \rightarrow$  spot price

$k \rightarrow$  risk free rate

For continuous compounding

$$F = Se^{kT}$$

$\rightarrow$  assumes no income, no storage cost

If  $F_0 > S_0 e^{rT}$ : Investor may buy the asset by borrowing an amount equal to  $S_0$  for a period of  $T$  at the risk free rate, and take a short position in forward contract. At the time of maturity, the assets will be delivered for a price of  $F$  and amount borrowed will be repaid by paying an amount equal to  $S_0 e^{rT}$  and the deal would result in a net profit of  $F_0 - S_0 e^{rT}$

If  $F_0 < S_0 e^{rT}$ : Investor would short the assets, invest the proceeds for the time period  $T$  at an interest rate  $r$  and long a forward contract. When the contract matures the asset would be purchased for a price of  $F$  and the short position in the asset would be closed out. This would result in a profit of  $S_0 e^{rT} - F_0$

### When an Investment Asset Provides a Known Income (preference Share, coupon bearing bonds)

$$F_0 = (S_0 - I)e^{rT}$$

where  $I$  is the present value of the income from investment during the life of future contract

Example:  $S_0 = \text{Rs. } 900$ ,  $I = 40e^{-0.03 \times 4/12} = \text{Rs. } 39.60$ ,  $r = 0.04$  and  $T = 0.75$

So:  $F_0 = (900 - 39.60)e^{0.04 \times 0.75} = \text{Rs. } 886.60$

If  $F_0 > (S_0 - I)e^{rT}$  an arbitrage can lock in a profit by buying the asset and shorting the a forward contract on the asset.

If  $F_0 < (S_0 - I)e^{rT}$  an arbitrage can lock in a profit by shorting the asset and taking a long position in a forward contract